D03 · Yield & Loss Reduction

03 · Six Sigma & DMAIC methodology

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1 WHAT IT IS

Six Sigma is a structured problem-solving methodology built around the DMAIC cycle (Define, Measure, Analyze, Improve, Control) applied to projects with measurable financial impact. The trained-belt structure (yellow, green, black, master black) defines methodology skill levels; the project portfolio defines the operational discipline. The methodology is only as effective as the project governance that selects, charters, executes, and verifies the savings — and that governance is what separates productive programs from belt-collection programs.

2 WHY IT MATTERS

Six Sigma is one of the most heavily invested-in quality methodologies of the last 30 years and the one with the most documented gap between certification and impact. **The dominant failure mode is investing in belt certifications without investing in project governance — producing trained people without finishing projects, and finishing projects without verifying the savings**. The reference text on the methodology quantifies the gap directly **(Pyzdek & Keller, 2018 — peer-reviewed, "The Six Sigma Handbook", 5th ed.)**: well-governed Six Sigma programs deliver **$200K–$500K of verified annual savings per certified black belt**; poorly-governed programs (typical certification mill) deliver under $50K per black belt and frequently produce projects that the finance team will not validate. The asymmetry is brutal: the same training investment, the same methodology, produces a 5–10x difference in business outcome depending on whether the project governance exists.

3 HOW IT IS DONE

1. **Charter projects against business problems with named financial owners, not against training certifications.** A project chartered as "the green belt needs to complete a DMAIC project to certify" is a project that will be selected for ease of completion, not for business impact. A project chartered as "reduce scrap on line 4 from $1.8M annually to $400K, with the operations director as financial sponsor" is a project tied to outcomes that matter. The charter discipline is the first and largest fork in the road.
2. **Run the Define phase with rigor — the wrong problem definition makes the rest of DMAIC useless.** The Define phase produces the project charter, the problem statement, the goal statement, the SIPOC, and the baseline business case. Most projects fail in the M-A-I-C phases because Define was rushed: the problem was too broad, the baseline was an estimate not a measurement, the goal was aspirational not committed. Time invested in Define returns 5–10x in execution velocity.
3. **Treat Measure as a verification step, not a homework assignment.** The Measure phase establishes the baseline that the Improve phase has to beat. If the baseline is wrong (measurement system noise, sampling bias, definition drift), the entire project is invalid. Most Six Sigma projects that fail finance verification fail because the Measure phase did not pass an MSA (measurement system analysis) — the project improved noise, not signal.
4. **Use Analyze to find the vital few drivers — not to demonstrate every statistical technique the belt learned in training.** The Analyze phase exists to identify which inputs (Xs) drive the output (Y) most strongly, so the Improve phase can attack the right Xs. The pathology is the belt who runs every tool in the training (DOE, ANOVA, regression, hypothesis tests) as a demonstration of skill rather than as a search for the vital few. Sharp Analyze produces 2–3 vital Xs; pathological Analyze produces a 40-page slide deck.
5. **Implement Improve with operational ownership transferred — not as a belt-led pilot that ends when the belt moves on.** The Improve phase is where the countermeasure gets implemented; the failure mode is implementation that depends on the belt's continued presence. The discipline is to transfer operational ownership of the countermeasure to the supervisor of the affected cell during Improve, with the belt as coach rather than owner, so that when the belt moves to the next project the gain stays.
6. **Verify Control with measurement that runs after the belt leaves — not with the closing presentation.** The Control phase exists to ensure the gain is sustained. Most programs run Control as a closing deliverable (control plan signed, SPC chart drawn) rather than as ongoing measurement. The correct discipline is 90-day, 180-day, and 365-day verification of the savings against the baseline, signed by finance, with regression-detection triggers that reopen the project if the gain decays.

Closure without sustained verification is closure that gets re-opened by next year's COQ analysis.

WORKED EXAMPLE — ILLUSTRATIVE

A medical-device manufacturer with FDA-regulated production launches a Six Sigma program over 30 months, certifying **47 green belts and 8 black belts** with classroom + project work. Two years in, finance audit reveals the impact picture: **52 projects chartered, 47 completed, 11% completed with financially verified savings, total verified annual benefit = $380K (~$8K per certified belt)** — well below the $200K/belt benchmark. Diagnosis: projects chartered for certification completion, not for business problems; Define phase rushed; Measure phase with no MSA; finance not in the project until closing; no sustained Control. The implementation: a 9-month relaunch installs project governance — operations director chairs a monthly project portfolio review; every charter requires finance sign-off on baseline and goal; mandatory MSA gate at Measure-end; finance validation at Control-end; 90/180/365-day verification cadence. Results 12 months after relaunch: **project completion rises from 11% to 84%, average verified savings per project rises from $73K to $310K, total annualized verified savings = $4.2M**. The training program did not change; the governance did. The same belts now produce 10x the business outcome.

IN PRACTICE

Verified savings with finance signature, or the project doesn't close

Well-governed programs deliver $200K–$500K verified annual savings per black belt; certification-mill programs deliver under $50K **(Pyzdek & Keller, 2018 — peer-reviewed, 5th ed.)**. The difference is the finance sign-off. Projects that finance won't validate didn't deliver.

Charter against business problems with named operational sponsors

The quality function supports the program; it does not own it. Operations supervisors sponsor projects day-to-day; the operations director chairs the portfolio review. Without operational sponsorship, projects optimize the methodology rather than the operation.

Run Define with the rigor that makes the rest of DMAIC work

A poorly defined problem produces 15 weeks of Measure–Analyze on the wrong question. The Define gate is where rigor pays off — problem statement, baseline, target, scope, financial impact — and the master black belt should reject Define charters that aren't ready.

Verify Control with measurement that runs after the belt leaves

Most project gains decay within 12 months because Control measurement stops the day the belt declares victory. The structural fix is to hand Control measurement to the cell supervisor on a permanent SOP, with quarterly verification by quality engineering for at least two years.

WHAT YOU WOULD USE

SIX SIGMA METHODOLOGY & PORTFOLIO

Minitab Statistical Software / JMP (SAS) — The enterprise statistical workbenches for DMAIC analysis — capability studies, MSA, hypothesis testing, DOE, regression — and the dominant tools in formal Six Sigma training curricula **(Minitab — vendor; SAS Institute — vendor)**.

Companion by Minitab / EngineRoom / iSixSigma — Project portfolio platforms purpose-built for Six Sigma program management — DMAIC-staged project workflow, savings tracking, belt registry, portfolio reporting; the operational layer between methodology and finance **(Minitab — vendor)**.

ASQ / Villanova / external training providers + internal master black belt — The training and certification supply — external providers for foundational belt training, internal master black belt for project coaching and program governance; the right structure when the methodology is being built in-house **(ASQ — vendor)**.

Smartsheet / Excel with disciplined templates + Minitab for analysis — For operations without a dedicated Six Sigma platform — DMAIC-templated project trackers in Smartsheet, statistical analysis in Minitab/JMP, monthly portfolio review chaired by operations leadership. Works when the project governance is real; fails when the tracker becomes the program.

4 SO WHAT

Six Sigma without project governance is a certification program with a savings narrative attached — and the narrative does not survive the finance audit. Charter against business problems with named financial owners, run Define with the rigor that makes the rest of DMAIC work, treat Measure as verification not homework, use Analyze to find the vital few not to demonstrate every tool, implement Improve with operational ownership transferred, and verify Control with measurement that runs after the belt leaves. The methodology is the procedural backbone; the project governance is what converts the methodology into business outcome — and the 5–10x impact difference between well-governed and belt-collection programs lives entirely in that conversion.

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